Preliminaries

X psd : $X \succeq 0$

$A \succeq B$: $(A-B) \succeq 0$ i.e. $A-B$ <u>PSD</u>.

"$A \cdot B$" $= \langle A, B \rangle = \sum_{i,j} A_{ij} B_{ij}$

$$\boxed{\begin{array}{l} A \succeq 0 \text{ all e-vals} \geq 0 \\ A \preceq 0 \text{ all e-vals} \leq 0 \\ A \succeq 0 \iff -A \preceq 0 \end{array}}$$

check : v a vector, vv^T is $n \times n$ matrix

$\langle vv^T, A \rangle = v^T A v$.

$A, B \succeq 0 \Rightarrow \underline{\langle A, B \rangle \geq 0}$

$A \succeq 0$, $A = \sum \overset{\geq 0}{\lambda_i} v_i v_i^T$

$\langle A, B \rangle = \langle \sum \lambda_i v_i v_i^T, B \rangle = \sum \lambda_i \underbrace{\langle v_i v_i^T, B \rangle}_{v_i^T B v_i} \geq 0.$

Semidefinite Programs

Two common formats of SDP

$\boxed{\text{I}}$

$$\boxed{\begin{aligned} \min:\ & \sum c_i x_i \\ \text{s.t.:}\ & \sum \underset{m\times m}{A_i} x_i \preceq \underset{m\times m}{B} \end{aligned}}$$

$\boxed{\text{II}}$

$$\begin{aligned} \min:\ & \langle \overset{n\times n}{C}, \overset{n\times n}{X} \rangle \\ \text{st:}\ & \langle A_i, X \rangle = b_i, \quad i=1,\dots,m \\ & X \succeq 0 \end{aligned}$$

For LPs:

$$\begin{aligned} \min:\ & \sum c_i x_i \\ \text{st:}\ & \sum \underset{m\times 1}{A_i} x_i \le \underset{m\times 1}{b} \end{aligned}$$

For LPs:

$$\begin{aligned} \min:\ & c^T x \\ \text{st:}\ & a_i^T x = b_i, \quad i=1,\dots,m \\ & x \ge 0 \end{aligned}$$

SDP as a Generalization of LP

LP

min: $\sum c_i x_i$

st: $\boxed{Ax \le b}$

$$\begin{bmatrix} | & & | \\ A_1 & \cdots & A_n \\ | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \sum A_i x_i$$

SDP $\supsetneq$ LP

SDP: min: $\sum c_i x_i$

st: $\sum \mathcal{A}_i x_i \preceq \mathcal{B}$

Want: $\mathcal{A}_i, \mathcal{B}$ – diagonal

Z diag. $Z \succeq 0 \iff Z_{ii} \geq 0 \ \forall i$

$$\mathcal{B} = \begin{pmatrix} \ddots & & 0 \\ & b & \\ 0 & & \ddots \end{pmatrix} = \mathrm{diag}(b)$$

Exercise: what are the $\mathcal{A}_i$'s ?

SDP duality

$$\left(B - \sum A_i x_i\right) \succeq 0$$

$$\begin{aligned} \min:\ & c^T x \\ \text{st:}\ & \sum \underbrace{A_i}_{m \times m} x_i \preceq \underbrace{B}_{m \times m} \end{aligned} \quad \longleftrightarrow$$

"punish" $\preceq 0$

$$\min_x:\ \max_{Z \succeq 0}:\ c^T x + \left\langle Z, \overbrace{B - \sum_i A_i x_i}^{\preceq 0} \right\rangle$$

$$\max_{Z \succeq 0}:\ \min_x:\ \sum_i \left(c_i - \langle Z, A_i \rangle\right) x_i + \langle Z, B \rangle$$

$$\begin{aligned} \max:\ & \langle Z, B \rangle \\ \text{st:}\ & \langle Z, A_i \rangle = c_i \quad \forall i \\ & Z \succeq 0 \end{aligned}$$